Curriculum Vitae

Taehun Cha
Ph.D. in Mathematics

Education

Ph.D. in Mathematics – Korea University (Advisor: Donghun Lee) Mar. 2022 – Aug. 2026

- Funded by National Research Foundation (NRF) of Korea
- Dissertation: *Two Faces of One Gram: Understanding Feature Learning with Virtual Covariance and Target Linearity*

M.F.E. in Financial Engineering – Korea University (Advisor: Donghun Lee) Mar. 2020 – Feb. 2022

- Funded by Financial Engineering Program
- Thesis: *Understanding the Yield Curve Shift with FOMC Statements: NLP Perspective*

B.A. in Sociology and Cultural Theory and Criticism – Yonsei University Mar. 2012 – Aug. 2019

- Minor in Applied Statistics

Work Experience

KT Corp., Research Intern Jul. 2023 – Aug. 2023

- Selected as an outstanding intern.

Publication

Mathematical Foundations of Artificial Intelligence

- **Taehun Cha**, Daniel Beaglehole, Adityanarayanan Radhakrishnan, and Donghun Lee. 2026. "The Weight Gram Matrix Captures Sequential Feature Linearization in Deep Networks." *The Fortieth Annual Conference on Neural Information Processing Systems* (**NeurIPS 2026**).
- **Taehun Cha** and Donghun Lee. 2025. "ABC3: Active Bayesian Causal Inference with Cohn Criteria in Randomized Experiments." In *The 39th Annual AAAI Conference on Artificial Intelligence* (**AAAI 2025, Oral**).
- **Taehun Cha** and Donghun Lee. 2025. "Curse of Smoothness in Functional Neural Networks." ***IEEE Signal Processing Letters***. DOI: 10.1109/LSP.2025.3624683.
- **Taehun Cha** and Donghun Lee. 2025. "Feature Learning as a Virtual Covariance Learning." *Optimization for Machine Learning Workshop at Neurips 2025* **(OPT@NeurIPS 2025)**.
- **Taehun Cha** and Donghun Lee. 2025. "Emergent Linear Separability of Unseen Data Points in Last-Layer Feature Space." *Workshop on High-dimensional Learning Dynamics at ICML 2025* **(HiLD@ICML 2025)**.

Natural Language Processing & AI Safety

- Chandler Smith, Marwa Abdulhai, (...), **Taehun Cha**, (...). 2025. "Evaluating Generalization Capabilities of LLM-Based Agents in Mixed-Motive Scenarios Using Concordia." *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track* (**NeurIPS 2025 D&B Track**).
- **Taehun Cha** and Donghun Lee. 2024. "Pre-trained Language Models Return Distinguishable Probability Distributions to Unfaithfully Hallucinated Texts." *Findings of the Association for Computational Linguistics: EMNLP 2024* (**EMNLP 2024 Findings**).
- **Taehun Cha** and Donghun Lee. 2024. "SentenceLDA: Discriminative and Robust Document Representation with Sentence Level Topic Model." In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics* (**EACL 2024, Oral**).

Artificial Intelligence for Scientific Discovery

- **Taehun Cha** and Donghun Lee. 2026. "Consequence-Guided Information Extraction for Predicting Central Bank Communication's Effect." ***Computational Economics***. DOI: 10.1007/s10614-025-11284-6. [Economics]
- **Taehun Cha** and Donghun Lee. 2024. "Evaluating Extrapolation Ability of Large Language Model in Chemical Domain." *Language + Molecules Workshop at ACL 2024* **(Lang+Moles@ACL 2024)**. [Chemistry]
- **Taehun Cha***, Jaeheun Jung*, and Donghun Lee. 2022. "Noun-MWP: Math Word Problems Meet Noun Answers." In *Proceedings of the 29th International Conference on Computational Linguistics* (**COLING 2022**). [Math Education]

Multidisciplinary Projects on Scientific Discovery

[Chemistry] **Extrapolative Chemical Property Prediction with LLM Chemical Knowledge**. 2024.

[Seismology] **Earthquake Detecting Algorithm from Multi-Observatory Data Using Deep Learning**. 2022.

[Math Education] **Large Scale Deep Learning Based Algorithm for Mathematics Problem Solving**. 2021.

Awards

Concordia Contest @ NeurIPS (2024) – First Place out of 197 participants, Cooperative AI Foundation, Google DeepMind, MIT, UC Berkeley, and UCL.

The 6th AI Grand Challenge (2023) – 7th Place, Ministry of Science and ICT.

Korean AI Competition (2022) – 4th Place, Ministry of Science and ICT.

The 5th AI Grand Challenge (2021), Ministry of Science and ICT.

Grant

Ph.D. Fellowship (2025), National Research Foundation of Korea (NRF). 25,000,000 KRW (∼20,000 USD)

Junior Fellow-Research Grant (2023), Korea University. 2,000,000 KRW (∼1,500 USD)

AI Grand Challenge (2021), Ministry of Science and ICT. 237,500,000 KRW (∼167,000 USD)

Talk and Presentation

Understanding Feature Learning 2: Target Linearity

- 2026 Korean Society for Industrial and Applied Mathematics, November 6, 2026.
- Yonsei University, September 30, 2026. (Host: Dohyun Kwon)
- National University of Singapore & Institute for Mathematical Sciences, July 01, 2026.
- University of California San Diego, June 08, 2026. (Host: Mikhail Belkin)

Understanding Feature Learning 1: Virtual Covariance

- Korea Institute For Advanced Study, January 05, 2026. (Poster)
- Korean Mathematical Society, October 22, 2025.

Cooperative AI: Focusing on the Concordia Contest, Cooperative AI Talk, AI Safety Asia, Online, April 10, 2025. https://www.youtube.com/watch?v=c4bSITHm6Ms

Extracting Financial Knowledge from FOMC Communications, Gangneung-Wonju National University, June 9, 2023.

Summer/Winter Research Program

- Annual Summer School on Mathematical Aspects of Data Science. National University of Singapore & Institute for Mathematical Sciences. 2026. (Travel grant recipient)
- KIAS Center for AI and Natural Sciences 2026 Winter Workshop. Korea Institute For Advanced Study. 2026.
- KIAS Lecture Series on Computational Neuroscience and AI. Korea Institute For Advanced Study. 2025.
- Machine Learning and Mathematics. Korea Institute For Advanced Study. 2025.
- NIMS Summer Tutorial on Artificial Intelligence. National Institute for Mathematical Sciences. 2024.

Teaching Experience

- Korea University Mathematics for Artificial Intelligence (2025 Fall), Teaching Fellow.

Academic Service

- Reviewer: NeurIPS, TMLR, ARR, OPT(W), AI4Research(W)